The Structure of Understanding: theory core

Sections 2.1–2.4 (Phase 0) with Phase 1 amendments

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Draft of 2026-10-23. No result in this document is evidence; planted-answer checks are estimator tests only.

Abstract

This document fixes the objects that the rest of the programme measures. It defines a concept inside a system as a variable of internal state that passes three tests (quotient, derivation, causal use), defines the concept graph and the equivalence it is defined up to, defines generality, principles, depth of understanding and structural memorization, and defines the three moves a new input can require (lower, sideways, diagonal) together with creation and false creation. Each definition carries a quantitative test and a proposed threshold. Each claim carries its status. Eight propositions have complete proofs in `proofs/` that have not yet been independently checked; every other claim is a conjecture with a named experiment. Nothing here is evidence; the evidence is the pre-registered programme in Sections 5–6 of the master prompt.

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1 Setting (master prompt §2.1)

Definition 1.1 (System, sites and states)**.** A *system* is a measurable map $f\colon \mathcal{X} \to \mathcal{Y}$ computed through a finite set of *sites* $\mathcal{S}$ partially ordered by a computation order $\prec$ (for a network: (layer, position, component) triples ordered by the forward pass; for a brain: (region, time window) pairs ordered by time). Each site s carries a state $h_s(x) \in \mathbb{R}^{d_s}$. We fix once and for all a linear extension of $\prec$, written $s_1 < s_2 < \cdots < s_{|\mathcal{S}|}$, and write $s < t$ for it. The input x is treated as the state of a source site s_0, and the output as the state of a sink site $s_\star$.

Definition 1.2 (Task distribution and behaviours)**.** A task distribution P on $\mathcal{X}$ is given. A *behaviour* is a pair $b = (P_b, r_b)$ of an input distribution $P_b \ll P$ and a readout $r_b\colon \mathcal{Y} \to \mathcal{R}_b$ into a finite set or a metric space (for a language model: the answer token of a task family; for an animal: the choice on a trial type). A *behaviour suite* is a finite family $\mathcal{B} = \{b_1, \dots, b_m\}$ with weights $w_b \geq 0$, $\sum_b w_b = 1$. For each b we fix a divergence d_b on the outputs relevant to r_b (total variation between output distributions, or 0/1 disagreement of r_b).

Definition 1.3 (Interventions and access levels)**.** For a site s, a *realiser* is a map $R\colon \mathbb{R}^{d_s} \to \mathbb{R}^k$ together with a right inverse that writes R-coordinates back into the state while leaving the complementary part fixed (a linear subspace with its orthogonal projection, as in distributed alignment search; or a set of sparse-dictionary features whose activations are swapped while the reconstruction error is kept). For inputs x, x' the *realiser patch* $f_{R \leftarrow x'}(x)$ is the output of the run on x in which the R-coordinates of $h_s(x)$ are replaced by those of $h_s(x')$ and all later computation is recomputed. Patches at several realisers are applied jointly. A result obtained with patches is at access level (C). At level (B) a patch is replaced by its first-order Jacobian estimate; at level (D) by the strongest perturbation the data allow (stimulation, optogenetics, or, failing that, decoding plus a natural-experiment contrast); at level (A) by counterfactual input edits. Every statement below is written for level (C); its weaker versions inherit the level of the instrument and must say so.

Remark 1.4 (Nulls attached to the setting). Every quantity defined below is computed on three systems in addition to f: an untrained copy f^0 with the same architecture (master prompt §1.4), a feature-shuffle null that keeps marginals, and random realisers of matched dimension and norm. A quantity earns a claim only through its contrast with these nulls.

2 Concept (master prompt §2.2)

A candidate concept is a triple (s, R, φ) of a site, a realiser at that site, and a *value map* $\varphi\colon \mathbb{R}^k \to V$ into a finite set or a metric space. Its value on an input is

$$C(x) = \varphi\big(R(h_s(x))\big).$$

Candidates are proposed by sparse dictionaries, by the null space of the pullback metric $g_s = J_s^\top J_s$ of the downstream map (master prompt §1.9), and by probes for known variables. Proposal confers nothing; only the three conditions below do.

Definition 2.1 (Partition of a concept)**.** The *partition* of C is $\pi_C = \{C^{-1}(v) : v \in V\}$, taken modulo P-null sets. For continuous V with metric ρ, fibres are replaced by δ-fibres $\{(x, x') : \rho(C(x), C(x')) \le \delta\}$ at a resolution δ fixed in pre-registration.

2.1 Condition 1: quotient

Definition 2.2 (Quotient defect and compression)**.** For a behaviour b, the *within-fibre swap effect* and the *unconditioned swap effect* are

$$A_b^{=}(C) = \mathbb{E}_{x,x' \sim P_b \,:\, C(x)=C(x')}\, d_b\big(f(x), f_{R \leftarrow x'}(x)\big), \qquad A_b(C) = \mathbb{E}_{x,x' \sim P_b}\, d_b\big(f(x), f_{R \leftarrow x'}(x)\big),$$

and the *quotient defect* is $\delta_Q(C) = \sum_b w_b\, A_b^{=}(C) \big/ \sum_b w_b\, A_b(C)$ over the behaviours with $A_b(C) > \eta$. The *compression ratio* is $\kappa(C) = H(C)/H(X)$, with $H(X) = \log N_{\text{eff}}$ for an empirical input set of effective size N_{eff}.

Quantitative test. C passes the quotient condition when (i) the realiser matters, $\sum_b w_b A_b(C) \ge \eta$ with η the 95th percentile of the same quantity for random realisers of matched dimension and norm; (ii) $\delta_Q(C) \le \varepsilon_Q$ with the upper end of its 95% bootstrap CI at most $2\varepsilon_Q$; (iii) $\delta_Q(C)$ is below the 5th percentile of δ_Q for random partitions of X with the same block-size profile as π_C (permutation test, Holm-corrected across candidates); (iv) $\kappa(C) \le \kappa_{\max}$ and $H(C) \ge h_{\min}$. **Proposed thresholds:** $\varepsilon_Q = 0.10$, $\kappa_{\max} = 0.25$, $h_{\min} = 0.5$ bit.

Remark 2.3. Condition 1 says that, inside its realiser, nothing but the value of C is visible to what follows. It does not say that the downstream computation depends on h_s only through C: other variables at the same site may be used too, and they are other candidates. Swaps that change the value of C are the subject of Condition 3. Chalupka, Perona and Eberhardt's causal coarsening of an input space is the closest prior construct; the difference is that the partition here is of internal states and is tested through realiser swaps.

2.2 Condition 2: derivation

Let $\mathcal{K}_{<s}$ be the set of concepts already accepted at sites strictly before s, together with the input variables (tokens, pixels, stimulus labels), which are the *input nodes*.

Definition 2.4 (Sufficient and minimal parent sets)**.** A set $S \subseteq \mathcal{K}_{<s}$ is *ε-sufficient* for C if $H(C \mid S) \le \varepsilon\, H(C)$, estimated as the held-out conditional cross-entropy of a sparse predictor $\hat{\psi}$ of C from S. It is *minimal* if no proper subset is 2ε-sufficient. The *parent set* $\mathrm{Pa}(C)$ is the minimal ε_D-sufficient set returned by sparse conditional prediction, and the *derivation map* is $\psi_C = \hat{\psi}$, so that $C \approx \psi_C(\mathrm{Pa}(C))$.

Quantitative test. C passes the derivation condition when (i) a minimal ε_D-sufficient set exists with $|\mathrm{Pa}(C)| \le k_{\max}$; (ii) the parent set is stable: mean pairwise Jaccard index at least $J_{\min}$ across at least five data resamples and, where they exist, five seeds; (iii) each edge is interventionally confirmed: patching the realiser of a parent C_i from x' into x moves C to ψ_C evaluated at the patched parent values on at least a fraction τ_E of pairs, and at least 0.2 above random realisers. **Proposed thresholds:** $\varepsilon_D = 0.05$, $k_{\max} = 8$, $J_{\min} = 0.8$, $\tau_E = 0.9$.

Proposition 2.5 (Derivation is refinement in the partition lattice; proved, `proofs/P1_partition_lattice.tex`)**.** *Order partitions of X by refinement, $\pi \le \pi'$ if every block of π lies inside a block of π', and write $\bigwedge$ for the common refinement (meet). In the exact case $\varepsilon_D = 0$ and for finite X, $C = \psi(C_1, \dots, C_k)$ for some map ψ if and only if $\bigwedge_i \pi_{C_i} \le \pi_C$. Consequently S is an exact parent set of C if and only if S is an inclusion-minimal subset of $\mathcal{K}_{<s}$ whose meet refines π_C.*

Measurable prediction. None by itself; it fixes what the estimator must recover and is the basis of the planted-answer check.
Experiment. Phase 1, planted system V1 (hand-built digit, column-sum, carry, result network): the estimated parent sets must equal the planted ones on every node at the pre-registered recovery threshold.

Proposition 2.6 (Uniqueness of the minimal parent set under positivity; proved, `proofs/P2_parent_uniqueness.tex`)**.** *Let $U = \mathcal{K}_{<s}$ be finite-valued and suppose the joint law of U under P gives positive probability to every combination of values in the product of the observed value sets of its members (positivity of the candidates; C itself may be a deterministic function of U). Then the inclusion-minimal set $S \subseteq U$ with $C \perp U \setminus S \mid S$ is unique.*

Measurable prediction. Where positivity holds, parent sets estimated on disjoint data halves coincide up to sampling error; where it fails, they may differ, and the difference is located on value combinations that never co-occur.
Experiment. Phase 1, planted systems V1 and V3; on V3 (components seen separately, never together) parent-set stability must drop exactly at the withheld combinations.

Remark 2.7 (Positivity is a real restriction)*.* Positivity fails whenever the candidate set contains a node together with a function of it, for example a node and its own parents, so it cannot be assumed for all of $\mathcal{K}_{<s}$. With $C = C_3 = C_1 \oplus C_2$ both $\{C_3\}$ and $\{C_1, C_2\}$ are minimal sufficient sets. Until a canonical candidate set is fixed (one option: the latest-site antichain that is sufficient), the programme reports all minimal parent sets and every downstream quantity as a range over the resulting graphs. Fixing the rule is a Phase 1 design decision, tested on V1 before any real model.

Proposition 2.8 (Derivation maps are unidentified on unseen combinations and identified by interchange; proved, `proofs/P3_unseen_combinations.tex`)**.** *(a) If a combination t of parent values has probability zero under P, then for any ψ that agrees with C P-almost surely there is a ψ' that also agrees with C P-almost surely and differs from ψ at t. (b) Suppose the parent realisers are* modular*: there exist inputs $x, x^{(1)}, \ldots, x^{(k)}$ such that the joint patch of the realiser of C_i from $x^{(i)}$ into the run on x produces parent values t and leaves the computation from the parents to C unchanged in form. Then the value the system assigns to C at t is observed directly, so the system's derivation map is identified at t by the interchange intervention.*

Measurable prediction. Observational fits cannot predict sideways behaviour; interchange tests on the new combination can. This is the formal content of the sideways prediction in §4.
Experiment. Phase 2–3, K3: on sideways inputs, the agreement between the interchange-identified ψ and the correct answer predicts correctness better than any observational fit of ψ.

2.3 Condition 3: causal use

Definition 2.9 (Interchange intervention accuracy)**.** Let D_C be the *downstream derivation* of C: the sub-graph of the current concept graph from C to the output, with its derivation maps, which defines a high-level model $F_{\mathcal{G}}$ in the sense of causal abstraction (Geiger, Lu, Icard and Potts). For a pair (x, x') with $C(x) \neq C(x')$ the predicted counterfactual output is $\hat{y}(x; C \leftarrow C(x')) = F_{\mathcal{G}}(x; \mathrm{do}(C = C(x')))$. The interchange intervention accuracy is

$$\mathrm{IIA}(C) = \sum_b w_b \, \mathbb{P}_{x,x' \sim P_b \,:\, C(x) \neq C(x')} \Big[r_b\big(f_{R \leftarrow x'}(x)\big) = r_b\big(\hat{y}(x; C \leftarrow C(x'))\big) \Big].$$

A graph-free variant replaces $\hat{y}$ by the system's output on a matched counterfactual input x^* that agrees with x on every other parent of the children of C and has $C(x^*) = C(x')$, where such an input exists in the sample. This variant breaks the circularity between Condition 3 and the graph on the first pass of discovery.

Quantitative test. C passes the causal-use condition when $\mathrm{IIA}(C) \geq \tau_{\mathrm{IIA}}$, the lower end of its 95% CI is above IIA of random realisers of matched dimension by at least Δ_{IIA}, and the same realiser in the untrained copy f^0 does not pass. A candidate whose probe decoding accuracy is at least 0.9 but that fails this test is recorded as *represented, not used*. **Proposed thresholds:** $\tau_{\mathrm{IIA}} = 0.90$, $\Delta_{\mathrm{IIA}} = 0.20$.

Definition 2.10 (Concept)**.** A *concept* of f (at access level (C)) is a candidate that passes Conditions 1, 2 and 3. Input nodes are concepts by convention and are exempt from Conditions 1 and 2.

2.4 The concept graph and its equivalence

Definition 2.11 (Concept graph)**.** Run discovery site by site in the order $<$. The concept graph $\mathcal{G}(f)$ has as nodes the concepts, taken modulo the equivalence below, and an edge $C_i \to C$ for each $C_i \in \mathrm{Pa}(C)$. A node's site is the earliest site at which a member of its class is accepted.

Definition 2.12 (Equivalence)**.** Two concepts C, C' are *equivalent* if $\pi_C = \pi_{C'}$ modulo P-null sets; approximately, if the normalised variation of information $\mathrm{VI}(\pi_C, \pi_{C'})/\log N_{\mathrm{eff}} \leq \varepsilon_{\mathrm{merge}}$. Two graphs are equivalent if there is a bijection of nodes that maps each node to an equivalent node and preserves edges. Equivalence absorbs (i) any invertible reparametrisation $u\colon V \to V'$ of a value map, since $\pi_{u \circ C} = \pi_C$, and (ii) the merging of redundant nodes that carry the same partition at different sites or in different realisers. **Proposed threshold:** $\varepsilon_{\mathrm{merge}} = 0.05$.

Proposition 2.13 (Invariance under reparametrisation; proved, `proofs/P4_reparametrisation.tex`)**.** *Every test statistic in Conditions 1–3 (δ_Q, κ, $H(C)$, $H(C \mid S)$,* IIA*) and every quantity in §3 (generality, grounding, memorization) is unchanged when a value map φ is replaced by $u \circ \varphi$ for an invertible u, provided the derivation maps of C and of its children are replaced by $u \circ \psi_C$ and $\psi_{C''} \circ (\dots, u^{-1}, \dots)$ respectively.*

Proposition 2.14 (Well-definedness in the exact case; proved, `proofs/P2_parent_uniqueness.tex`, Corollary)**.** *Assume (a) the exact case $\varepsilon_Q = \varepsilon_D = 0$ and $\varepsilon_{\mathrm{merge}} = 0$, finite X and finite value sets; (b) positivity of the candidate parents at every site (Proposition 2.6); (c) the set of accepted candidates is fixed. Then $\mathcal{G}(f)$ is a directed acyclic graph that is unique up to the equivalence of Definition 2.12, and it does not depend on the choice of value maps within each equivalence class.*

Conjecture 2.15 (Stability of the approximate graph)**.** *Suppose the exact-case assumptions hold for some limiting system and that every accepted concept clears each threshold with margin $\mu > 0$ and every rejected candidate fails some threshold by margin μ. Then for $\varepsilon_Q, \varepsilon_D, \varepsilon_{\mathrm{merge}}$ small relative to μ and for sample size $n \to \infty$, the estimated graph equals the exact-case graph up to equivalence with probability tending to one.*

Status. Conjecture. Obstruction to a proof: the sparse-prediction estimator of $\mathrm{Pa}(C)$ needs a margin condition analogous to the beta-min condition of sparse regression, and its form for conditional entropy is not yet stated.

Measurable prediction. Graphs estimated on disjoint data halves, and on seeds of the same training recipe, agree up to equivalence at a rate that rises with n.
Experiment. Phase 1 (V1–V4), recovery vs sample size; Phase 2 (K1), split-half graph agreement on Llama, Qwen, Gemma and OLMo.

Remark 2.16 (Dependence on the linear extension). Acyclicity holds by construction because edges respect the fixed linear extension $<$. Whether the graph depends on the choice of extension when sites are incomparable under $\prec$ (parallel attention heads, parallel brain regions) is open; it is recorded as Conjecture 2.15's companion and tested by recomputing the graph under two extensions.

2.5 Relation to Formal Concept Analysis

A formal context is a triple (G, M, I) of objects, attributes and an incidence relation; with $A' = \{m : \forall g \in A,\ gIm\}$ and $B' = \{g : \forall m \in B,\ gIm\}$, a formal concept is a pair (A, B) with $A' = B$ and $B' = A$ (Wille 1982; Ganter and Wille 1999). We attach to a site the *downstream context* $\mathbb{K}_s = (X, M_s, I)$, whose attributes M_s are binary downstream readouts $m\colon X \to \{0, 1\}$ (thresholded behaviour readouts and thresholded later-site features), with xIm iff $m(x) = 1$. Say that C is an *exact quotient for* M_s if every $m \in M_s$ is constant on every block of π_C.

Proposition 2.17 (Quotient condition and formal concepts; proved, `proofs/P5_fca.tex`)**.** *Let C be an exact quotient for M_s. (a) Every extent of $\mathbb{K}_s$ is a union of blocks of π_C. (b) For every x, the block $[x]_C$ is contained in the object-concept extent $\{x\}''$. (c) If M_s is closed under complement ($m \in M_s \Rightarrow 1 - m \in M_s$, the nominal scaling of FCA), then the object-concept extents $\{\{x\}'' : x \in X\}$ are exactly the blocks of the kernel partition π_{M_s} of $x \mapsto (m(x))_{m \in M_s}$; $\pi_C \le \pi_{M_s}$; and C is the coarsest exact quotient for M_s if and only if its blocks are the object-concept extents.*

Conjecture 2.18 (Approximate version)**.** *If C has quotient defect $\delta_Q(C) \le \varepsilon$ with respect to a behaviour suite whose thresholded readouts form M_s, then every extent of $\mathbb{K}_s$ differs from a union of π_C-blocks on a set of P-probability at most $c\,\varepsilon$, with c depending only on the thresholding margin of the readouts.*

Status. Conjecture. Obstruction: δ_Q is a ratio of averaged divergences, so it bounds the mass of disagreeing pairs only after the readouts are thresholded with a margin.
Measurable prediction. On a model with a known graph, FCA extents of the downstream context are, to within $c\,\varepsilon$, unions of the fibres of the accepted concepts; on the untrained copy they are not.
Experiment. Phase 1, V1 and V2; reported as a check on Condition 1, never as evidence for a claim.

Remark 2.19 (What the FCA relation does and does not say). Proposition 2.17 shows that the quotient condition is a one-site, interventional form of FCA's closure: concepts that pass Condition 1 induce saturated extents, and under nominal scaling the coarsest such concept is FCA's object-concept partition. FCA by itself has neither a derivation relation (its lattice order is subsumption of extents, not computation of one variable from others) nor a causal-use test, so Conditions 2 and 3 are not consequences of it.

3 Generality, principles and depth (master prompt §2.3)

Definition 3.1 (Active derivation of an input)**.** For an input x in the support of P_b, node C is *active on* x if its realiser swap moves the readout on a majority of partners:

$$a_C(x) = \mathbf{1}\Big[\mathbb{P}_{x' \sim P_b:\, C(x') \neq C(x)}\big(r_b(f_{R \leftarrow x'}(x)) \neq r_b(f(x))\big) \geq \alpha\Big].$$

The active derivation $D(x)$ is the sub-graph of $\mathcal{G}(f)$ induced by the active nodes that are ancestors of the output node, together with the input nodes. **Proposed threshold:** $\alpha = 0.5$.

Definition 3.2 (Generality)**.** The *weighted generality* of a node is the behaviour-weighted mass of inputs on whose derivation it is active,

$$g(C) = \sum_{b \in \mathcal{B}} w_b \, \mathbb{E}_{x \sim P_b} \, a_C(x) \in [0, 1],$$

and the *count generality* is $g_{\#}(C) = \#\{b : \mathbb{E}_{P_b} a_C \geq \alpha\}$. The *null-corrected* generality is $g(C) - q_{95}$, where q_{95} is the 95th percentile of g over the nodes of the untrained-copy graph (or, if that graph is empty, over random realisers of matched dimension).

Definition 3.3 (Principle)**.** A *principle at level* τ is a node that is not an input node and has $g(C) \geq \tau$ and $g(C) > q_{95}$. Write Π_τ for the set of principles. **Proposed thresholds:** $\tau \in \{0.25, 0.5\}$ pre-registered as two levels; all results are also reported as curves in τ.

Definition 3.4 (Depth of understanding)**.** Let $\mathcal{P}(x)$ be the set of directed paths in $D(x)$ from an input node to the output.

(i) *Thresholded depth.* The behaviour is *grounded at* τ *on* x if $\Pi_\tau \cap D(x)$ is a vertex cut separating the input nodes from the output in $D(x)$.

(ii) *Structural grounding level.* $\lambda(x) = \min_{p \in \mathcal{P}(x)} \max_{C \in p} g(C)$, the bottleneck value of the most weakly grounded path. Then x is grounded at τ if and only if $\lambda(x) \geq \tau$ (Proposition 3.5).

(iii) *Interventional grounding share.* With Π_τ patched from x into the run on x',

$$\gamma_\tau(x) = 1 - \frac{\mathbb{E}_{x' \sim P_b} \, d_b\big(f(x), f_{\Pi_\tau \leftarrow x}(x')\big)}{\mathbb{E}_{x' \sim P_b} \, d_b\big(f(x), f(x')\big)},$$

the share of the input's effect on the output that passes through principles ($\gamma = 1$: the output on x is fixed once the principles are fixed; $\gamma = 0$: principles carry nothing).

(iv) *Depth of a behaviour.* $\Lambda_\tau(b) = \mathbb{P}_{x \sim P_b}(\lambda(x) \geq \tau)$ (thresholded), $\bar{\lambda}(b) = \mathbb{E}_{P_b} \lambda(x)$ and $\bar{\gamma}_\tau(b) = \mathbb{E}_{P_b} \gamma_\tau(x)$ (continuous), and the τ-integrated score $\int_0^1 \bar{\gamma}_\tau(b) \, d\tau$.

Proposition 3.5 (Bottleneck characterisation; proved, `proofs/P6_bottleneck.tex`)**.** *In any finite DAG with node weights g, the nodes with $g \geq \tau$ form a vertex cut between the input nodes and the output if and only if $\min_p \max_{C \in p} g(C) \geq \tau$. Hence (i) and (ii) of Definition 3.4 agree, and $\lambda(x)$ is computable by a bottleneck path search in time linear in the number of edges after sorting.*

Conjecture 3.6 (Structural grounding implies interventional grounding)**.** *Assume $\mathcal{G}(f)$ is* complete *on x (every computational path from the input to the output passes through a node of $D(x)$), the quotient and interchange conditions hold exactly, and realisers are modular. If $\lambda(x) \geq \tau$ then $\gamma_\tau(x) = 1$. With approximations, $1 - \gamma_\tau(x)$ is bounded by a constant times $\varepsilon_Q + (1 - \tau_{\mathrm{IIA}})$ plus the completeness defect.*

Status. Conjecture. The exact case has a proof sketch (patching the cut fixes every downstream value by the quotient and interchange conditions); the sketch is not complete because completeness of $\mathcal{G}(f)$ is not itself measurable without a residual term, which has not been defined.
Measurable prediction. $\bar{\lambda}(b)$ and $\bar{\gamma}_\tau(b)$ are strongly rank-correlated across behaviours, and their disagreement is largest where the graph's mediation share is lowest.
Experiment. Phase 1 planted systems; Phase 2 K1 (arithmetic, where the completeness defect can be measured against the known algorithm).

Definition 3.7 (Structural memorization)**.** On input x, the behaviour is *memorized at* τ if no principle is active on its derivation, $\mu(x) = \max_{C \in D(x) \setminus \text{inputs}} g(C) < \tau$ (with $\mu(x) = 0$ if $D(x)$ has no non-input node). The behaviour b is memorized if $\mathbb{P}_{P_b}(\mu(x) < \tau) \geq 1 - \varepsilon_M$, and *grounded* if $\Lambda_\tau(b) \geq 1 - \varepsilon_M$; otherwise it is *partly grounded*. The verdict uses only training-distribution inputs and interventions; it uses no held-out data. **Proposed threshold:** $\varepsilon_M = 0.10$.

Proposition 3.8 (Order of the scores; proved, `proofs/P6_bottleneck.tex`)**.** *$\lambda(x) \leq \mu(x)$ for every x. Hence an input that is grounded at $\tau > 0$ is not memorized at τ, and, provided $\varepsilon_M < 1/2$, the three verdicts (grounded, partly grounded, memorized) are mutually exclusive for a behaviour.*

Conjecture 3.9 (Depth predicts generalization; the K2 kill test)**.** *Across held-out inputs of a behaviour, per-input error is a decreasing function of $\lambda(x)$ and of $\gamma_\tau(x)$, and these scores predict per-input error with higher AUROC than Bernardi et al.'s cross-condition generalization score, max-softmax, entropy, margin and sharpness, at the pre-registered margin.*

Status. Conjecture. This is the programme's kill test; if it fails the programme stops (master prompt §6.2, K2).
Measurable prediction. AUROC of λ for per-input error exceeds every baseline in §6.4 with non-overlapping 95% CIs, on held-out arithmetic, syntactic agreement and a diagnosis model.
Experiment. Phase 2, K2.

Conjecture 3.10 (Memorization verdict predicts held-out failure)**.** *A behaviour judged memorized by Definition 3.7 has held-out accuracy no better than the best input-free baseline, and a behaviour judged grounded has held-out accuracy within the training-accuracy CI, provided the specimen gate (training accuracy at ceiling) is met.*

Measurable prediction. The structural verdict, computed without held-out data, separates leakage-trained from clean models and planted-memorized from planted-rule subsets.
Experiment. Phase 1 V2 (estimator test only); Phase 4 K4 and K9 (claims).

Remark 3.11 (Relation to Bernardi et al.)*.* Proposition 2.7.1 of the master prompt (cross-condition generalization as the depth-one case of depth) is attempted in Phase 1, not here. The expected form is a statement about a single-node graph with a linear value map and observational data, with the causal-use term only able to lower the score.

4 The three moves and creation (master prompt §2.4)

Definition 4.1 (Training range)**.** For a node C with finite V, its *value range* $\mathrm{Rg}(C)$ is the set of values taken on at least $n_{\min}$ training inputs. For continuous V, $\mathrm{Rg}(C)$ is the set of values whose k-nearest-neighbour distance to training values is below the 99th percentile of the same distance among training values. The *combination range* of C is $\mathrm{Rg}(\mathrm{Pa}(C))$, the set of parent-value tuples seen on at least $n_{\min}$ training inputs. A realiser state *has a value* if it lies within the 99th-percentile distance of the realiser states of some fibre. **Proposed thresholds:** $n_{\min} = 5$, $k = 10$.

Definition 4.2 (Lower, sideways and diagonal)**.** Let x^* be an input not seen in training, with active derivation $D(x^*)$.

(i) x^* is *lower* if, for every node $C \in D(x^*)$, the realiser state has a value, $C(x^*) \in \mathrm{Rg}(C)$, and the parent tuple $\mathrm{Pa}(C)(x^*) \in \mathrm{Rg}(\mathrm{Pa}(C))$.

(ii) x^* is *sideways* if every node value is in range but at least one parent tuple is not.

(iii) x^* is *diagonal* if the derivation needs a concept that is not in $\mathcal{G}(f)$. Operationally: some realiser state on $D(x^*)$ has no value, or the graph's mediation share $\gamma_{\mathcal{G}}(x^*)$ (Definition 3.4(iii) with all non-input nodes patched) is below $1-\varepsilon_{\mathrm{cov}}$, so part of the input's effect on the output passes through no node. In benchmarks, diagonal inputs are also labelled by construction (the task requires a variable that is not a function of any node).

Proposed threshold: $\varepsilon_{\mathrm{cov}} = 0.2$. The operational classifier is $r.\mathrm{move}(x)$; its agreement with construction labels is tested at a pre-registered AUROC.

Conjecture 4.3 (Lower inputs are answered correctly)**.** *If x^* is lower and the behaviour is grounded, the system's error on x^* is at most its training error plus a term that vanishes as $\varepsilon_Q, \varepsilon_D, 1-\tau_{\mathrm{IIA}} \to 0$.*

Status. Conjecture. In the exact case it reduces to the statement that every node's value on x^* is computed by maps that were exercised on the same arguments in training; the obstruction is that "exercised in training" does not imply "correct" unless training error is zero on those arguments.

Conjecture 4.4 (Sideways inputs: correct iff interchange passes)**.** *If x^* is sideways, the system answers correctly if and only if, for each unseen parent tuple t on $D(x^*)$, the interchange intervention that assembles t from training inputs (Proposition 2.8(b)) yields the value of C prescribed by the task's ground-truth derivation.*

Conjecture 4.5 (Diagonal inputs fail without creation)**.** *If x^* is diagonal, the system fails on it unless a creation event (Definition 4.6) occurs under exposure to x^*'s problem.*

Measurable prediction. For Conjectures 4.3–4.5 jointly: accuracy orders lower > sideways > diagonal; within sideways, the interchange verdict predicts correctness at a pre-registered AUROC; within diagonal, success occurs only with a valid new node.
Experiment. Phase 3, K3 (moves); Phase 7, K13 (diagonal with creation). Estimator check on planted system V3 in Phase 1.

Definition 4.6 (Creation)**.** Let $e_1, e_2, \ldots$ be an exposure sequence to a new problem (in-context examples for a language model, training steps for a network, sessions for an animal), measured in units u (tokens, gradient steps, sessions), and let $\mathcal{G}_t$ be the concept graph after exposure t, recomputed with the same estimator and thresholds. A *valid new node* at t is a node C^* of $\mathcal{G}_t$ such that

(i) C^* passes Conditions 1–3 in the exposed system;

(ii) C^* is not equivalent to any node of $\mathcal{G}_0$ (VI above $\varepsilon_{\mathrm{merge}}$ to every node);

(iii) its parents are nodes of $\mathcal{G}_0$, or new nodes that are themselves valid, so that it is derived from existing nodes;

(iv) it does not appear, at the same exposure count, under exposure to a matched control problem that does not require it.

The *creation rate* is the number of valid new nodes per unit u; the *creation reach* of C^* is its generality $g_t(C^*)$ over the original suite $\mathcal{B}$ together with the new behaviour.

Definition 4.7 (False creation)**.** A node that appears under exposure is a *false creation* if (a) it is decodable but not used (passes probe decoding, fails Condition 3); (b) it is used but not derived (passes Condition 3, fails Condition 2 with respect to $\mathcal{G}_0$), which makes it a new leaf, i.e. memorization of the new problem; (c) it is a relabelling (equivalent to a node of $\mathcal{G}_0$, including an invertible reparametrisation of one); or (d) it is exposure-generic (it appears equally under the control problem). Every reported creation count lists the candidates rejected under (a)–(d).

Conjecture 4.8 (Creation predicts success on a diagonal problem)**.** *Across systems and problems, success on a diagonal problem after exposure is predicted by the appearance of at least one valid new node whose parents lie on the derivation of the problem, and the creation rate is positively associated with the speed of success; false creations do not predict success.*

Status. Conjecture; the creation tests run last because they are the most likely to fail.
Measurable prediction. Logistic regression of success on (valid-node indicator) has a positive coefficient with CI excluding zero across the K13 problem set; the same regression on false-creation counts does not.
Experiment. Phase 7, K13; estimator check on planted system V5 in Phase 1.

5 Phase 1 amendments M1–M8 (estimator design decisions, 2026-10-23)

Planted-answer development (V1–V5, development seeds only) exposed places where the Phase 0 definitions either fail on a planted answer or leave an estimator choice open. Each amendment below was fixed before the frozen validation seeds were run (`prereg/PHASE_1.md`) and is part of the estimator definition from then on. None was chosen on real data.

(M1) **Interventional direct-effect parents.** With parents chosen by observational minimal sufficiency (Definition 2.4), a planted rule-plus-exceptions network (V2) gets the output's parent set $\{a, b\}$ (an input lookup), so its rule node never lies on any derivation and every input is scored memorized. The parent set is therefore the set of pool nodes P at earlier sites whose *direct* effect on C (swap P from partners with a different P value, every pool node at a site strictly between held at its clean value) exceeds the effect of a within-fibre swap of P by at least $\theta_{\text{dir}} = 0.05$. Sufficiency ($H(C \mid \text{Pa}(C)) \leq \varepsilon_D H(C)$), $|\text{Pa}(C)| \leq k_{\max}$ and resample stability remain required. Observational minimal sufficient sets are still enumerated and reported as alternatives (Remark 2.7).

(M2) **Edge activity in the active derivation.** $D_b(x)$ keeps the active nodes of Definition 3.1 and adds *active edges*: $P \to C$ is active on x if the direct-effect swap of P changes C (or r_b) on at least α times the fraction of the same partners whose own C value differs from $C(x)$. A node lies on the derivation if it is on an active-edge path from an input to the output. Generality (Definition 3.2) counts inputs on whose derivation the node lies in this sense.

(M3) **Leaf rule.** In λ and μ a node contributes $g(C)$ on x only if $C(x) \in \text{Rg}(C)$ (Definition 4.1); otherwise it is a leaf on x and contributes 0. This is what makes "each input is effectively its own leaf" (master prompt §2.3.4) measurable: an exception node whose value on x is shared by fewer than $n_{\min}$ training inputs does not ground x.

(M4) **Canonical realisers.** (i) In residual architectures the probe realiser of a variable at site s_ℓ is decoded from what the block wrote, $h_\ell - h_{\ell-1}$ (ridge decoder column space; $h_{\ell-1}$ is the same token position at the previous exposed layer); at s_0 from the state. A probe realiser is proposed only if its value map decodes its label with 2-fold held-out accuracy ≥ 0.9 (proposal confers nothing; the filter only saves interventions). (ii) A candidate whose realiser lies inside a larger candidate (or input realiser) at the same site whose partition refines it is a *coarsening* and is not a separate node. (iii) A candidate covered by the span of two or more smaller, pairwise *separable* (state-level modular) quotient-passing candidates is a *composite* and is not a node. (iv) Equivalent candidates are merged; interventions use the realiser at the node's earliest surviving site with the lowest quotient defect; members at other sites are bookkeeping.

(M5) **The discovery fixed point.** Derivation failures are resolved first, earliest site first; a node that fails derivation is suspended (it is re-tested on every pass and reactivated if it passes once the pool changes). Causal-use failures are resolved latest site first. Until the graph is stable, causal use is judged on effective pairs (those for which $F_{\mathcal{G}}$ predicts a change) and a node with too few effective pairs is pending; the final pass applies the full test (all-pairs IIA $\geq \tau_{\text{IIA}}$, effective IIA $\geq \tau_{\text{IIA}}$, margin Δ_{IIA} over norm-matched random realisers).

(M6) **Nulls and calibrations.** Random realisers are norm-matched (each patch delta rescaled to the norm of the candidate's own delta). Condition 1(i) is per behaviour: the realiser must matter more than the 95th percentile of norm-matched random realisers for at least one behaviour. $\kappa_{\max}$ is calibrated from the proposed 0.25 to 0.6 (the proposed value rejects every non-trivial node of a two-input system); $H(X)$ is the sum of the input nodes' entropies.

(M7) **Output realisers.** A candidate whose partition is equivalent to a behaviour's output partition is a realiser of that output node, not a derivation step, exactly as input-equivalent candidates are realisers of input nodes. Without this, the representation of the answer itself lies on every derivation with generality equal to its behaviour's mass, and every input of a single-behaviour suite is "grounded" through its own answer (found on V2). The output node's parents are then the parents of that representation.

(M8) **Interventional identification of derivation maps (Proposition 2.8(b)).** When a child's parents include a node and a function of it (positivity fails, Remark 2.7), every interchange on the node produces a parent tuple never seen observationally, and $F_{\mathcal{G}}$ is undefined there. Derivation maps ψ_C are therefore fitted on the observational tuples together with the tuples produced by the direct-effect interventions of M1 (the patched parent at its source value, the others clean or held). IIA is evaluated on fresh pairs; agreement at a tuple reached both ways is the modularity condition of Proposition 2.8(b).

Remark 5.1 (What the amendments change in the statuses). Propositions 2.5–2.17 concern the exact observational objects and are unchanged. Proposition 3.5 applies to the active-edge DAG of (M2) with the node weights of (M3) verbatim.

6 Propositions 2.7.1–2.7.3 of the master prompt

Proposition 6.1 (CCGP and depth one; proved, `proofs/P7_ccgp.tex`)**.** *Bernardi et al.'s CCGP is not the depth-one case of depth: there are systems with perfect depth-one grounding and CCGP* $\leq 1/2$*, and systems with CCGP* $= 1$ *for a causally inert variable. With the decoder fixed to the system's own readout and causal use added, readout-aligned CCGP* $= 1$ *iff the depth-one candidate passes the exact quotient and causal-use tests.*

Proposition 6.2 (Grounded behaviours generalize to lower inputs; proved under (A1)–(A3), `proofs/P8_grounded_lower.tex`)**.** *If a set of principles is a vertex cut of the derivation on a new input, the nodes below the cut are value-local, and the cut's joint value was taken by a training input with the same derivation below the cut, the system's output on the new input equals its output on that training input; it is correct if the target factors through the principle values.*

Proposition 6.3 (Memorization and sideways generalization; proved, `proofs/P9_memorized_sideways.tex`)**.** *(a) A behaviour memorized at τ can generalize sideways (a private compositional node used by one behaviour of a large suite). (b) If the behaviour is leaf-memorized (every node active on a training input has an empty training range), every new input is diagonal.*

7 Summary of statuses

Item	Status	Test
Prop. 2.5 derivation as refinement	proved, unchecked	V1
Prop. 2.6 parent uniqueness	proved, unchecked	V1, V3
Prop. 2.8 unseen combinations	proved, unchecked	K3
Prop. 2.13 reparametrisation	proved, unchecked	unit/property tests
Prop. 2.14 well-definedness (exact)	proved, unchecked	V1–V4
Prop. 2.17 FCA relation	proved, unchecked	V1, V2
Prop. 3.5 bottleneck	proved, unchecked	unit tests
Prop. 3.8 order of scores	proved, unchecked	unit tests
Conj. 2.15 stability	conjecture	V1–V4, K1
Conj. 2.18 approximate FCA	conjecture	V1, V2
Conj. 3.6 structural vs interventional	conjecture	V1, K1
Conj. 3.9 depth predicts error (kill)	conjecture	K2
Conj. 3.10 memorization verdict	conjecture	V2, K4, K9
Conj. 4.3–4.5 three moves	conjecture	V3, K3, K13
Conj. 4.8 creation	conjecture	V5, K13
Prop. 6.1 CCGP vs depth one	proved, unchecked	K2 (disagreement analysis)
Prop. 6.2 grounded ⇒ lower correct	proved under (A1)–(A3), unchecked	K3, K6

Item	Status	Test
Prop. 6.3 memorized vs sideways	proved (counterexample + leaf form), unchecked	V2, K3

Proposed thresholds (to be fixed in pre-registration)

Symbol	Proposed value	Role
ε_Q	0.10	quotient defect
$\kappa_{\max}$, $h_{\min}$	0.25 (calibrated to 0.6, M6), 0.5 bit	compression, non-triviality
θ_{dir}	0.05	direct-effect parent (M1)
ε_D, $k_{\max}$	0.05, 8	parent sufficiency, parent count
$J_{\min}$, τ_E	0.8, 0.9	parent stability, edge interchange
τ_{IIA}, Δ_{IIA}	0.90, 0.20	causal use, margin over null
$\varepsilon_{\text{merge}}$	0.05	equivalence
α	0.5	per-input activity
τ	0.25 and 0.5	principle levels
ε_M	0.10	memorization verdict
$n_{\min}$, k	5, 10	training range
ε_{cov}	0.2	diagonal coverage

A Proofs

The proofs below are complete as written but have not been checked line by line by a second reader; until that check is recorded they support propositions, not theorems.

A.1 Proof of Proposition 2.5 (derivation as refinement)

Setting. X is finite and every $x \in X$ has $P(x) > 0$ (restrict X to the support of P; this is what "modulo P-null sets" means for finite X). Concepts $C_1, \dots, C_k, C$ are maps from X into finite sets.

Claim. There is a map ψ with $C(x) = \psi(C_1(x), \dots, C_k(x))$ for all x if and only if $\bigwedge_i \pi_{C_i} \leq \pi_C$.

Proof. The blocks of the meet $\bigwedge_i \pi_{C_i}$ are the non-empty sets $\{x : C_i(x) = t_i \ \forall i\}$, one for each tuple $t = (t_1, \dots, t_k)$ in the image T of $x \mapsto (C_1(x), \dots, C_k(x))$.

($\Rightarrow$) Let x, x' lie in one block of the meet. Then $C_i(x) = C_i(x')$ for every i, hence $C(x) = \psi(C_1(x), \dots) = \psi(C_1(x'), \dots) = C(x')$, so x, x' lie in one block of π_C. Every block of the meet therefore lies inside a block of π_C.

($\Leftarrow$) For $t \in T$ choose any x_t with tuple t and set $\psi(t) = C(x_t)$; for $t \notin T$ set $\psi(t)$ arbitrarily. The choice does not matter: all inputs with tuple t form one block of the meet, which lies in one block of π_C, so they share one value of C. Then for every x with tuple t, $\psi(t) = C(x_t) = C(x)$. □

Corollary. An exact parent set (an inclusion-minimal S with $H(C \mid S) = 0$) is exactly an inclusion-minimal $S \subseteq \mathcal{K}_{<s}$ with $\bigwedge_{C' \in S} \pi_{C'} \leq \pi_C$, because for finite X with full support $H(C \mid S) = 0$ holds if and only if C is a function of S on X.

A.2 Proof of Proposition 2.6 (uniqueness of the minimal parent set) and of Proposition 2.14

Setting. All variables are finite-valued functions of $X \sim P$. $U = \{U_1, \ldots, U_n\}$ is the candidate parent set; for $A \subseteq U$ write A also for the tuple of its members. *Positivity*: for every combination $u = (u_1, \ldots, u_n)$ in $\prod_j \mathcal{V}_j$, where $\mathcal{V}_j$ is the set of values U_j takes with positive probability, $P(U = u) > 0$. Positivity of U implies positivity of every sub-collection, since a marginal of a positive law on a product set is positive on the product of the marginal value sets. A set $S \subseteq U$ is a *blanket* for C if $C \perp U \setminus S \mid S$.

Lemma (intersection property with a deterministic target). Let Y, W, Z be disjoint sub-collections of U and assume positivity. If $C \perp Y \mid (Z, W)$ and $C \perp W \mid (Z, Y)$, then $C \perp (Y, W) \mid Z$. No positivity is required of C.

Proof. Fix values z, y, w. By positivity $P(y, w, z) > 0$ for every y and w in their value sets, so all conditionals below are defined. The first hypothesis gives $P(c \mid y, z, w) = P(c \mid z, w)$ and the second gives $P(c \mid y, z, w) = P(c \mid z, y)$. Hence $P(c \mid z, w) = P(c \mid z, y)$ for all y, w. The left side does not depend on y and the right side does not depend on w, so both equal a function $q(c, z)$ only. Then $P(c \mid z) = \sum_{y,w} P(c \mid y, z, w) P(y, w \mid z) = q(c, z) \sum_{y,w} P(y, w \mid z) = q(c, z)$, and therefore $P(c \mid y, z, w) = q(c, z) = P(c \mid z)$ for all y, w, which is $C \perp (Y, W) \mid Z$. □

Proof of Proposition 2.6. Let S_1, S_2 be blankets. Put $Z = S_1 \cap S_2$, $A = S_1 \setminus S_2$, $B = S_2 \setminus S_1$ and $R = U \setminus (S_1 \cup S_2)$; these are disjoint and cover U. Since $S_1 = Z \cup A$ is a blanket, $C \perp (B, R) \mid (Z, A)$. Since $S_2 = Z \cup B$ is a blanket, $C \perp (A, R) \mid (Z, B)$, and by weak union $C \perp A \mid (Z, B, R)$. Apply the Lemma with $Y = A$ and $W = (B, R)$: the two hypotheses are $C \perp A \mid (Z, B, R)$ and $C \perp (B, R) \mid (Z, A)$, both established. Hence $C \perp (A, B, R) \mid Z$, i.e. $Z = S_1 \cap S_2$ is a blanket. Now let S_1, S_2 be inclusion-minimal blankets. Then $S_1 \cap S_2$ is a blanket contained in both, so by minimality $S_1 = S_1 \cap S_2 = S_2$. □

Link to Definition 2.4. If C is a function of U (the exact case of Condition 2, where some subset of U is 0-sufficient), then for any $S \subseteq U$, S is a blanket iff $H(C \mid S) = 0$. Indeed, if $H(C \mid S) = 0$ then C is a function of S and is independent of everything given S; conversely, if $C \perp U \setminus S \mid S$ then $H(C \mid S) = H(C \mid S, U \setminus S) = H(C \mid U) = 0$. So under positivity the exact parent set of Definition 2.4 is unique.

Proof of Proposition 2.14. Nodes are equivalence classes of accepted candidates, i.e. distinct partitions of X; since the accepted set is fixed (assumption (c)), the node set is determined. Each class is assigned the earliest site of its members under the fixed linear extension, which is determined. For each node C the candidate parents form a positive collection by (b), so its exact parent set is unique by the result just proved; hence the edge set is determined. Every edge goes from a node whose site is strictly earlier to a node whose site is later in a fixed linear order, so a directed cycle would require a site strictly earlier than itself; the graph is acyclic. Finally, by Proposition 2.13, replacing a value map by an invertible reparametrisation leaves the partition, hence the node, and all conditional-independence statements, hence the edges, unchanged. □

Caveat recorded with the proof. Positivity fails whenever the candidate set contains two members one of which is a function of the other (for example a node together with its own parents), and equivalent candidates violate it trivially; merging (Definition 2.12) is therefore a precondition, not a convenience. When positivity fails, distinct minimal parent sets can exist (for $C = C_3 = C_1 \oplus C_2$ both $\{C_3\}$ and $\{C_1, C_2\}$ are minimal). The programme then reports the set of all minimal parent sets and every downstream quantity as a range over the resulting graphs; the choice of a canonical candidate set (for example, the latest-site antichain that is sufficient) is an open design decision for Phase 1 and is tested on planted system V1.

A.3 Proof of Proposition 2.8 (unseen combinations)

(a) Let $T = (C_1, \ldots, C_k)$ be the parent tuple and t a value with $P(T = t) = 0$, and let ψ satisfy $C = \psi(T)$ P-a.s. Assume $|V| \geq 2$ (otherwise C is constant and fails $h_{\min}$). Choose $v \neq \psi(t)$ in V and define $\psi'(t) = v$ and $\psi' = \psi$ elsewhere. Then $P(\psi'(T) \neq \psi(T)) \leq P(T = t) = 0$, so $C = \psi'(T)$ P-a.s., while $\psi'(t) \neq \psi(t)$. No observational quantity distinguishes ψ from ψ'. □

(b) This part is definitional and is stated so that its assumption is explicit. *Modularity* means: there is a map Ψ_{sys}, defined on the parent-value tuples reachable by joint realiser patches, such that in every run (natural or patched) the value of C computed by the system equals Ψ_{sys} of the parent values present in that run. Under modularity, a joint patch that produces parent values t yields a run in which C takes the value $\Psi_{\text{sys}}(t)$, which is read off by $\varphi \circ R$ at the site of C. Hence $\Psi_{\text{sys}}(t)$ is observed, and it agrees with ψ wherever $P(T = \cdot) > 0$. □

Remark. Modularity is an assumption about the system, not a consequence of Conditions 1–3; it is the assumption under which the interchange test is evidence about the unseen combination, and it is itself checked by testing that the patch does not alter C's other parents (Definition 2.4, test (iii)).

A.4 Proof of Proposition 2.13 (reparametrisation)

Let $u\colon V \to V'$ be a bijection and $\tilde{C} = u \circ C$, with the same site and realiser. The statement concerns population quantities, not the behaviour of a particular finite-sample estimator.

(1) *Fibre events.* $\tilde{C}(x) = \tilde{C}(x')$ iff $C(x) = C(x')$ because u is injective, so $\pi_{\tilde{C}} = \pi_C$. The realiser patch $f_{R \leftarrow x'}(x)$ does not involve φ. Hence $A_b^{=}$, A_b and δ_Q are unchanged.

(2) *Entropies.* $H(u(C)) = H(C)$ and $H(u(C) \mid S) = H(C \mid S)$ for bijective u; likewise $H(C \mid S)$ is unchanged when a member of S is reparametrised, since conditioning depends only on the generated partition. So κ, $h_{\min}$, sufficiency and minimality, and hence $\mathrm{Pa}(\cdot)$, are unchanged.

(3) *Interchange.* With $\psi_{\tilde{C}} = u \circ \psi_C$ and each child's map $\tilde{\psi}_{C''}$ obtained from $\psi_{C''}$ by precomposing u^{-1} in the C argument, the high-level model satisfies $\tilde{F}_{\mathcal{G}}(x; \mathrm{do}(\tilde{C} = u(v))) = F_{\mathcal{G}}(x; \mathrm{do}(C = v))$ for every v, by direct substitution. The pairs with $\tilde{C}(x) \neq \tilde{C}(x')$ are the pairs with $C(x) \neq C(x')$, and the system-side patch is unchanged. So IIA is unchanged.

(4) *Generality and depth.* $a_C(x)$ depends only on the fibre events of (1) and on realiser patches, so g, $g_{\#}$, Π_τ, $D(x)$, λ, μ and γ_τ are unchanged.

(5) *Ranges.* For finite V, $\mathrm{Rg}(\tilde{C}) = u(\mathrm{Rg}(C))$ and $\tilde{C}(x^*) \in \mathrm{Rg}(\tilde{C})$ iff $C(x^*) \in \mathrm{Rg}(C)$.

□

Limit of the statement. For a continuous value space, the δ-fibres of Definition 2.2 and the nearest-neighbour range of Definition 4.1 are preserved by isometries of (V, ρ) but not by general homeomorphisms. The invariance claim for continuous V is therefore made only for isometries, and the choice of metric on V is part of the pre-registered specification of a concept.

A.5 Proof of Proposition 2.17 (quotient condition and Formal Concept Analysis)

Setting. $\mathbb{K}_s = (X, M_s, I)$ with X finite, M_s a set of maps $X \to \{0, 1\}$ and $xIm \iff m(x) = 1$. For $A \subseteq X$, $A' = \{m : \forall x \in A,\ m(x) = 1\}$; for $B \subseteq M_s$, $B' = \{x : \forall m \in B,\ m(x) = 1\}$. Extents

are the sets B' for $B \subseteq M_s$ (equivalently, the sets A with $A'' = A$). C is an exact quotient for M_s: each m is constant on each block of π_C.

(a) Let $E = B'$ be an extent and $x \in E$, and let $y \in [x]_C$. For each $m \in B$, $m(y) = m(x) = 1$, so $y \in B' = E$. Hence E contains, with each of its points, the whole π_C-block of that point, i.e. E is a union of blocks. □

(b) $\{x\}'' = \{x\}''$ is the extent of the attribute set $\{x\}'$; it contains x, so by (a) it contains $[x]_C$. □

(c) Assume M_s is closed under complement. Let $y \in \{x\}''$, so $m(x) = 1 \Rightarrow m(y) = 1$ for all $m \in M_s$. If $m(x) = 0$ then $(1-m)(x) = 1$ with $1-m \in M_s$, so $(1-m)(y) = 1$, i.e. $m(y) = 0$. Hence $m(y) = m(x)$ for every m, so y lies in the kernel block $[x]_{M_s}$. Conversely, if $y \in [x]_{M_s}$ then y has every attribute x has, so $y \in \{x\}''$. Thus $\{x\}'' = [x]_{M_s}$, and the object-concept extents are exactly the blocks of π_{M_s}. Since each m is constant on π_C-blocks, two points in one π_C-block have the same attribute vector, so $\pi_C \leq \pi_{M_s}$: every exact quotient refines π_{M_s}. The map $x \mapsto (m(x))_{m \in M_s}$ is itself an exact quotient with partition π_{M_s}, which is therefore the coarsest. Hence C is the coarsest exact quotient iff $\pi_C = \pi_{M_s}$, iff the blocks of π_C are the object-concept extents. □

Remark. Without closure under complement, (c) fails: with a single attribute m and $m(x) = 0$, $\{x\}' = \emptyset$ and $\{x\}'' = X$, which is strictly larger than the kernel block of x whenever m is not identically zero. This is why the downstream context is built with nominal scaling.

A.6 Proofs of Proposition 3.5 and Proposition 3.8 (bottleneck characterisation and order of scores)

Conventions. D is a finite DAG with a set I of input nodes, an output node o, and weights $g \geq 0$ on internal nodes (neither input nor output). For a path p from I to o, $\max_{C \in p} g(C)$ is taken over its internal nodes and is 0 if it has none. $\mathcal{P}$ is the set of such paths. If $\mathcal{P} = \emptyset$ we set $\lambda = 0$ (an output disconnected from the inputs is not grounded); μ is the maximum of g over internal nodes of D, and 0 if there are none. A set K of internal nodes is a vertex cut if every $p \in \mathcal{P}$ contains a node of K. For $\tau > 0$ let $K_\tau = \{C : g(C) \geq \tau\}$.

Proposition 3.5. Let $\tau > 0$ and $\mathcal{P} \neq \emptyset$. K_τ is a vertex cut iff every $p \in \mathcal{P}$ contains an internal node with $g \geq \tau$, iff $\max_p g \geq \tau$ for every p, iff $\min_{p \in \mathcal{P}} \max_{C \in p} g(C) \geq \tau$ (the minimum over a finite non-empty set is attained). Computation: process nodes in topological order and set $M(v) = 0$ for $v \in I$ and $M(v) = \min_{u \to v} \max\big(M(u),\, g(u)\mathbf{1}[u \notin I]\big)$ otherwise. By induction on the topological order, $M(v)$ is the minimum over paths from I to v of the maximum weight of the internal nodes before v; so $\lambda = M(o)$, computed with one pass over the edges. □

Proposition 3.8. If $\mathcal{P} = \emptyset$ then $\lambda = 0 \leq \mu$. Otherwise, for any $p \in \mathcal{P}$, $\max_{C \in p} g(C) \leq \mu$ because the internal nodes of p are internal nodes of D; hence $\lambda = \min_p \max_{C \in p} g \leq \mu$. If $\lambda(x) \geq \tau > 0$ then $\mu(x) \geq \tau$, so x is not memorized at τ. For a behaviour: $\{\lambda \geq \tau\} \subseteq \{\mu \geq \tau\}$, so $\mathbb{P}(\mu < \tau) \leq 1 - \Lambda_\tau(b)$. If b is grounded, $\Lambda_\tau(b) \geq 1 - \varepsilon_M$, hence $\mathbb{P}(\mu < \tau) \leq \varepsilon_M$, which is $< 1 - \varepsilon_M$ whenever $\varepsilon_M < 1/2$; so b is not memorized. Partly grounded is defined as neither, so the three verdicts are mutually exclusive when $\varepsilon_M < 1/2$. □

A.7 Proposition 2.7.1 of the master prompt: CCGP and depth one

Setting. A single site with noiseless condition means: a finite set of conditions K, a trial of condition k has state $h = \mu_k \in \mathbb{R}^d$, and conditions are equally weighted. A dichotomy $C \colon K \to \{0, 1\}$. The behaviour b reads out $y(h) = \mathbf{1}[w^\top h > \theta]$ for a fixed $w \neq 0$, θ. Bernardi et al.'s CCGP for C with m training conditions per side is the mean, over all choices S of m conditions from each side, of the accuracy on $K \setminus S$ of the max-margin linear classifier trained on $\{\mu_k : k \in S\}$. The

readout-aligned CCGP replaces that classifier by the system's own readout y. The depth-one graph is $K \to C \to y$ with C realised by $R = \mathrm{span}(w)$ and value map $\varphi(z) = \mathbf{1}[z > \theta/\|w\|]$ on $z = \hat{w}^\top h$.

Proposition P7. (a) *Depth-one grounding without CCGP.* There is a system in which y reports C on every condition (so the candidate passes the quotient test with $\delta_Q = 0$, the causal-use test with IIA $= 1$, and every input has $\lambda(x) = g(C) = 1$), while CCGP with $m = 1$ is at most $1/2$. (b) *CCGP without a concept.* There is a system with CCGP $= 1$ for a dichotomy C' whose realiser does not matter for any behaviour, so C' fails Condition 1(i) and contributes 0 to every depth score. (c) *Aligned case.* Readout-aligned CCGP equals 1 for every m if and only if $y(\mu_k) = C(k)$ for all k, which holds if and only if the depth-one candidate passes the exact quotient and causal-use tests; then $\lambda(x) = g(C) = 1$ on every input. Hence Bernardi's CCGP is the depth-one case of the depth measure only after the decoder is fixed to the system's own readout direction and causal use is added; without these two changes neither score bounds the other.

Proof. (a) Take $d = 2$, $C = 1$ conditions $\mu_1 = (1, 0)$, $\mu_2 = (-3, 10)$; $C = 0$ conditions $\mu_3 = (-1, 0)$, $\mu_4 = (3, -10)$; readout $w = (1, \frac{1}{2})$, $\theta = 0$. Then $w^\top \mu = 1, 2, -1, -2$, so $y = C$ on all four. The realiser swap of $z = \hat{w}^\top h$ between two trials with the same C leaves y unchanged (the swapped coordinate stays on the same side of the threshold), so $\delta_Q = 0$; a swap between trials with different C sets z to the partner's value and moves y to the partner's C, which is what the graph $C \to y$ predicts, so IIA $= 1$; C is active on every trial and lies on its only path, so $g(C) = 1 = \lambda(x)$. CCGP with $m = 1$ averages four training pairs. The max-margin classifier for two points is the perpendicular bisector. $\{\mu_1\}$ vs $\{\mu_3\}$: direction $(1, 0)$; it classifies μ_2 (-3) as 0 and μ_4 (3) as 1: accuracy 0. $\{\mu_2\}$ vs $\{\mu_4\}$: direction $(-6, 20)$ through the origin; μ_1 scores -6 (wrong), μ_3 scores 6 (wrong): accuracy 0. $\{\mu_1\}$ vs $\{\mu_4\}$: direction $\mu_1 - \mu_4 = (-2, 10)$, midpoint $(2, -5)$, offset $(-2, 10) \cdot (2, -5) = -54$; μ_2 scores $106 + 54 > 0$ (right), μ_3 scores $2 + 54 > 0$ (wrong): accuracy $1/2$. $\{\mu_2\}$ vs $\{\mu_3\}$ is symmetric: accuracy $1/2$. So CCGP $= (0 + 0 + \frac{1}{2} + \frac{1}{2})/4 = 1/4 \leq 1/2$. (b) Take $d = 2$, $\mu_k = (C'(k), \epsilon_k)$ with arbitrary small ϵ_k and a behaviour that reads only the second coordinate. Every training split yields a classifier whose direction is dominated by the first coordinate, and all held-out conditions are classified correctly (both sides are separated by $x_1 = \frac{1}{2}$ with margin $\frac{1}{2}$ against $|\epsilon|$ small), so CCGP $= 1$. Swapping the first coordinate never changes any readout, so $A_b(C') = 0$ for every behaviour and Condition 1(i) fails. (c) With the decoder fixed to y, the accuracy on held-out conditions is the fraction of $k \in K \setminus S$ with $y(\mu_k) = C(k)$. It is 1 for every S iff $y(\mu_k) = C(k)$ for all k (every condition is held out by some S). If $y(\mu_k) = C(k)$ for all k, the swap argument of (a) gives $\delta_Q = 0$ and IIA $= 1$, and $\lambda = g = 1$. Conversely if $y(\mu_k) \neq C(k)$ for some k, a swap from a trial of condition k into a trial of the other side moves y to $y(\mu_k)$, not to $C(k)$, so IIA < 1. □

Measurable prediction. On real models, CCGP and depth disagree in two identifiable ways: variables that are geometrically abstract but unused (high CCGP, failing Condition 1 or 3) and variables that are used but not parallel across conditions (low CCGP, high depth). The K2 contrast reports the per-input agreement of the two scores and the share of disagreements of each kind; that is what decides whether depth carries information CCGP lacks.

A.8 Proposition 2.7.2 of the master prompt: grounded behaviours and lower inputs

Setting and assumptions. Fix a behaviour b, its concept graph $\mathcal{G}$ and a set of principles $\Pi \subseteq \Pi_\tau$. Let T be the training inputs and x^* a new input. Write Desc(Π) for the nodes of $D_b(x^*)$ that are strictly downstream of Π on a path to the output node o (including o).

(A1) *Grounding on x^*.* Π is a vertex cut of $D_b(x^*)$ between the input nodes and o.

(A2) *Value locality downstream of the cut.* For every $C \in \mathrm{Desc}(\Pi)$ and every $x \in T \cup \{x^*\}$, $C(x) = \psi_C(\mathrm{Pa}_x(C)(x))$, where $\mathrm{Pa}_x(C)$ are the parents of C that are active on x, and ψ_C is a fixed function (the system's derivation map, which the exact derivation and interchange conditions identify on the parent tuples seen in training).

(A3) x^* *is lower for* Π. There is a training input $x_t \in T$ with $D_b(x_t) = D_b(x^*)$ on $\mathrm{Desc}(\Pi) \cup \Pi$ and $\Pi(x_t) = \Pi(x^*)$ (the joint tuple of principle values is in range).

Proposition P8. Under (A1)–(A3), $f_b(x^*) = f_b(x_t)$. Consequently, if training error is zero on x_t and the task's target factors through the principle values ($\mathrm{target}(x) = \rho(\Pi(x))$ for some ρ on $T \cup \{x^*\}$), the system answers x^* correctly.

Proof. Order $\mathrm{Desc}(\Pi)$ topologically. Claim: every $C \in \mathrm{Desc}(\Pi)$ has $C(x^*) = C(x_t)$. Take C and suppose the claim holds for all earlier members. Each active parent P of C on x^* lies on an active path to o through C. If P were neither in Π nor in $\mathrm{Desc}(\Pi)$, the active path from an input to P (which exists because $P \in D_b(x^*)$ lies on an input-to-output path) followed by $P \to C \to \cdots \to o$ would avoid Π: it cannot meet Π after C because C is already downstream of Π and the graph is acyclic, and it does not meet Π before P by the choice of P. That contradicts (A1). So every active parent of C is in Π or is an earlier member of $\mathrm{Desc}(\Pi)$; by (A3) and the induction hypothesis its value on x^* equals its value on x_t, and by (A3) the active parent sets agree. By (A2), $C(x^*) = \psi_C(\cdot) = C(x_t)$. Applying the claim to $C = o$ gives $f_b(x^*) = f_b(x_t)$. If $f_b(x_t)$ is correct and the target factors through Π, then $\mathrm{target}(x^*) = \rho(\Pi(x^*)) = \rho(\Pi(x_t)) = \mathrm{target}(x_t) = f_b(x_t) = f_b(x^*)$. □

What the assumptions cost. (A2) is the substantive one: it says the nodes below the principles are computed from their parents' values only, also on x^*. It is measurable on x^* by within-fibre swaps and by edge interventions from training partners (the interchange identification of Proposition 2.8(b)). (A3) is the definition of a lower input, restricted to the cut and below it; nodes upstream of the cut may be out of range. The factorisation of the target is a property of the task and the principles, not of the system. It fails exactly when a principle is a shortcut (a high-generality node that predicts the target on the training distribution only), and the proposition then predicts confident errors on lower inputs, the K6 shortcut signature.

Measurable prediction. Among held-out inputs that are lower for the principle cut, errors occur only where (A2) fails on the input (detected by interventions on that input) or where the target does not factor through the principles; errors on lower inputs with (A2) verified are therefore concentrated in behaviours whose principles are known shortcuts. Tested in K3 (moves) and K6 (shortcuts).

A.9 Proposition 2.7.3 of the master prompt: memorization and sideways generalization

The statement "a memorized behaviour has no sideways generalization" is false for memorization as defined (Definition 3.7); it is true for a stronger, leaf-level form.

Proposition P9. (a) *Counterexample.* For every $\tau > 0$ there is a system and a behaviour b that is memorized at τ, has a non-empty set of sideways inputs, and answers every sideways input correctly. (b) *Leaf-level memorization.* Say b is *leaf-memorized* if every non-input node C of $\mathcal{G}$ that is active on some training input of b has an empty training range, $\mathrm{Rg}(C) = \emptyset$ (no value of C is taken on $n_{\min}$ or more distinct training inputs). Then no new input is lower or sideways for b: every new input whose derivation contains a non-input node is diagonal, and so is every new input whose output is not mediated by the graph.

Proof. (a) Let $m > 1/\tau$ behaviours $b_1, \ldots, b_m$ with equal weight $1/m < \tau$. Behaviour b_1 computes $C = (A + B) \bmod q$ through a private node C used by no other behaviour, from nodes A, B computed from disjoint inputs, and reads $y = C$; behaviours $b_2, \ldots, b_m$ read other inputs through other private nodes. Training withholds a set W of (A, B) combinations while every value of A and of B is seen ($q \geq 3$, $|W|$ small). The block that computes C is the full addition table (the compositional variant of planted system V3). Every node on the derivation of b_1 (A, B, C) is used only by b_1, so $g \leq 1/m < \tau$, $\mu(x) < \tau$ on every input, and b_1 is memorized at τ. An input with $(A, B) \in W$ has all node values in range and the parent tuple of C unseen, so it is sideways; the system writes $(A + B) \bmod q$ there, which is correct. (b) Let x^* be new. If $D_b(x^*)$ has a non-input node C, then $C(x^*)$ is a value of C and $\mathrm{Rg}(C) = \emptyset$, so $C(x^*) \notin \mathrm{Rg}(C)$; by Definition 4.2 x^* is neither lower nor sideways, hence diagonal (operationally, the out-of-range value is reported). If $D_b(x^*)$ has no non-input node, the output is not mediated by the graph, which is diagonal by the operational rule of Definition 4.2(iii) (graph mediation share $0 < 1 - \varepsilon_{\mathrm{cov}}$). □

Consequences. The structural memorization verdict is a statement about reuse across the behaviour suite, not about compositionality inside one behaviour: (a) shows it can label a compositional but unshared computation as memorized. That is a design property of the definition, and it fixes what the K4/K9 memorization claims may say: a model judged memorized is predicted to have no *shared* principles, not to fail on every sideways input. The leaf-level form (b) is the one that predicts failure, through Conjecture 4.5.

Measurable prediction. (i) On systems trained with a planted memorized subset (V2 and its trained variant), inputs whose derivation is leaf-memorized are classified diagonal by r.move when perturbed to unseen neighbours, and fail. (ii) The counterexample family exists in real suites: behaviours judged memorized at τ whose private nodes pass the interchange test on unseen tuples generalize sideways; K3 reports how often.

References

Full, verified citations are in `docs/novelty.md`. Those used here: R. Wille (1982), Restructuring lattice theory; B. Ganter and R. Wille (1999), Formal Concept Analysis; A. Geiger, H. Lu, T. Icard and C. Potts (2021), Causal abstractions of neural networks; A. Geiger et al. (2025), Causal abstraction: a theoretical foundation for mechanistic interpretability; K. Chalupka, P. Perona and F. Eberhardt (2015), Visual causal feature learning; S. Bernardi et al. (2020), The geometry of abstraction in the hippocampus and prefrontal cortex; J. Pearl (1988), Probabilistic Reasoning in Intelligent Systems (Markov boundaries and the intersection property).